Class - 10

Inferential Statistics

DIFFERENCE

Aspect	Descriptive	inferential
Purpose	Summarizes and organizes data	Makes predictions or generalizations about a population
Focus	Deals with **what the data shows**	Deals with **what we can conclude beyond the data**
Data Used	Works with the entire dataset (population or sample)	Works with a sample to infer about the population
Techniques	Mean, median, mode, variance, SD, correlation, graphs, distributions	Sampling, hypothesis testing, confidence intervals, regression, statistical tests
Certainty	Provides definite results for given data	Involves uncertainty; results are probabilistic
Example	"The average height of students in this class is 165 cm."	"Based on this sample, the average height of all students in the school is about 165 cm (±2 cm)."

Probability

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Probability is the likelihood of an event happening, measured between 0 (impossible) and 1 (certain).

Probability = Likelihood

So basically probability is nothing but the events that can happen/occur.

Probability = Number of events occur

Probability

Example:

Let's take a example of a coin.

Their are two faces of coin one is head and the another one is tails.

Head **Tail**

If we flip the coin we either get head or tails. In this the events that are happening are either getting the head or tails.

Probability

So the probability of getting the heads and the tails is half

$$\text{Probability} = \frac{\text{Desired Outcome}}{\text{Total number of outcome}}$$

Probability of getting head = $\frac{1}{2}$

Probability of getting tail = $\frac{1}{2}$

Probability

Lets take an another example of a dice. It consists number from 1 to 6. So what is the probability of getting 6?

So as we know that every number on dice appear 1 times when it rolls ,So

Probability of getting 6 = $\frac{1}{6}$

What is the probability of getting a even number = $\frac{3}{6}$

What is the probability of getting a odd number = $\frac{3}{6}$